

1. You are provided with a dataset in a folder named **"Attendance Lists"**, which contains 36 individual excel workbooks of attendance list data for of our recent concluded training.
2. Use Microsoft Excel (or Google Sheets) to perform all computations, visualizations, and dashboard creation.
3. Show all working formulas clearly in each cell where calculations are made.
4. Save your completed work as **"YourName_FinalExam1.xlsx"** and submit it before the deadline.
5. Each of you must work independently. Collaboration or file sharing will result in disqualification.
6. The exam consists of four sections, **Data Preparation, KPI Computation, Dashboard Design, and Insight Interpretation.**
7. The total marks are 100.

SECTION A: DATA CLEANING AND PREPARATION (20 Marks)

You are required to prepare the dataset for analysis.

1. Inspect all the individual workbooks and identify any data quality issues (e.g., missing or invalid time values, inconsistent day names).
2. Combine all the 36 workbooks and create one single table of truth to use for analysis and create a new column **Duration (Hours)** that calculates the total time each participant spent in a session.
3. Create all other helper columns **that you think** will help you answer all the questions in your analysis.
4. Compute the **total hours** and **average duration per participant** for each Day.
5. Compute the **total hours attended per student** (summing up all Duration Hours by name).

SECTION B: KPI COMPUTATION (40 Marks)

Using your cleaned dataset, calculate the following Key Performance Indicators (KPIs). Make sure that you have included all other helper columns to be able to answer these questions properly. You will notice that each session is planned to take a maximum of **2 Hours** so that is our target for each session.

1. **Total Sessions in the entire course.** This is a static number of all sessions earlier indicated by each individual session.

2. **Total Overall Unique Students.** This is the count of unique students who attended all the 36 sessions.
3. **Overall Attendance Rate.** This tells us how actively the whole class participated on average across all sessions. It answers: *"Out of all possible student attendances, what percentage actually attended?"*
4. **Average attendance per session.** This tells you the average number of students who attended each session, it helps you know how big your audience typically was per day.
5. **Average Hours per session (all students combined).** This shows how much total classroom time was consumed on average during each session. If students stay longer in sessions, this value increases.
6. **Planned hours across all sessions.** This represents how many total hours the entire course was designed to last if every session ran fully. It's the **"planned time"** the benchmark against which we measure real attendance hours.
7. **Course completion Rate by hours.** The **Course Completion Rate (by Hours)** measures how many students actually *attended at least 80% of the total training time* (not just number of sessions).
8. **Individual attendance rate.** This shows how consistent each student was in showing up for sessions. It answers: *"Out of all 36 sessions, how many did this person attend?"*
9. **Individual Hours % of planned.** This compares how long each student actually spent in class versus how long the entire course was planned to run. It's a measure of total engagement time.
10. **Quality Index (Attendance Quality Score).** This KPI blends **attendance frequency** and **attendance duration** into one measure. It rewards students who not only attended many sessions but also stayed long enough in each session they attended.
11. **First Week Retention Rate.** Shows how many students remained engaged during the first week of training (Day1–Day7). It measures initial commitment and onboarding success.
12. **Mid-Course Retention Rate.** Tells you how many students were still participating halfway through the course (Day16–Day20). It shows if engagement was maintained after the start.
13. **Final Week Retention Rate.** Shows how many students stayed active towards the end (last 5 sessions). It reflects completion discipline and consistency.
14. **Volatility Index.** This shows how much attendance numbers fluctuated from day to day. High volatility means attendance was unstable; low volatility means it was steady.
15. **Session to Session Decline Rate.** Tells how much attendance dropped from the first to the last session, in percentage points. It's the simplest way to measure engagement decay.
16. **High Performers count (students who qualify for certificate).** It shows how many students demonstrated excellent attendance (both in frequency and duration). It identifies your most committed participants.
17. **At Risk Student Count.** Shows how many students are at risk of failing or disengagement due to poor attendance or very low hours.
18. **Drop Off students count.** These are students who started strong but stopped attending toward the ending.

SECTION C: DASHBOARD DESIGN (25 Marks)

Design a clean, interactive dashboard summarizing all the computed metrics.

Additional Charts and Visuals

1. **Chart 1.** Attendance Count by Day (showing trend). This Shows how class attendance changed from Day1 to Day36 and helps spot drop-offs or spikes in attendance.
2. **Chart 2:** Rolling 5-Day Average. Smooths short-term ups and downs in attendance. Shows the overall engagement pattern, e.g., steady decline after Day 20.
3. **Chart 3:** Top 10 Students by Quality Index. Identifies the most consistent and longest-attending students. Highlights role models or potential peer mentors.
4. **Chart 4:** Pie or Donut chart showing attendance segments (High Performers, Moderate, At-Risk).
5. Add slicers for filtering by **Day** and **Participant**.

SECTION D: INSIGHT AND INTERPRETATION (15 Marks)

Answer the following based on your findings:

1. What trend do you observe in attendance over time?
2. Which session had the highest and lowest attendance?
3. Identify three students with the best attendance and explain your reasoning.
4. Explain what might have caused the drop-off pattern observed.
5. Suggest two strategies that could improve attendance consistency.

BONUS (10 Marks)

Add a dynamic leaderboard that ranks students based on Quality Index using formulas or a pivot table.